

ML Scan Parameters

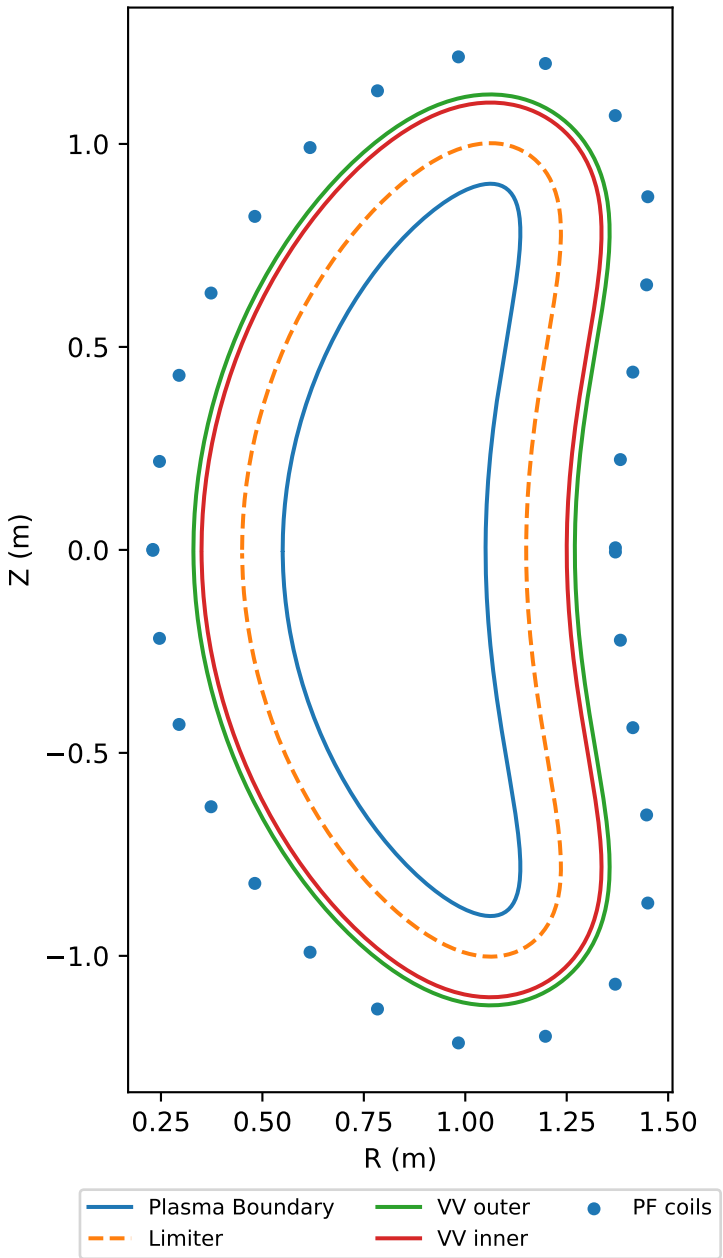
C-shape plasma $(r,z) = (A + B*\sin t + C*\cos 2t, F + G*\cos t - H*\sin 2t)$

A=0.9400 B=0.2500 C=0.1400
G=0.9000 H=0.0300 F=0.0
R0=0.8000 m kappa_target=3.600 delta=-0.030
B0=1.8 T Ip=1.00 MA

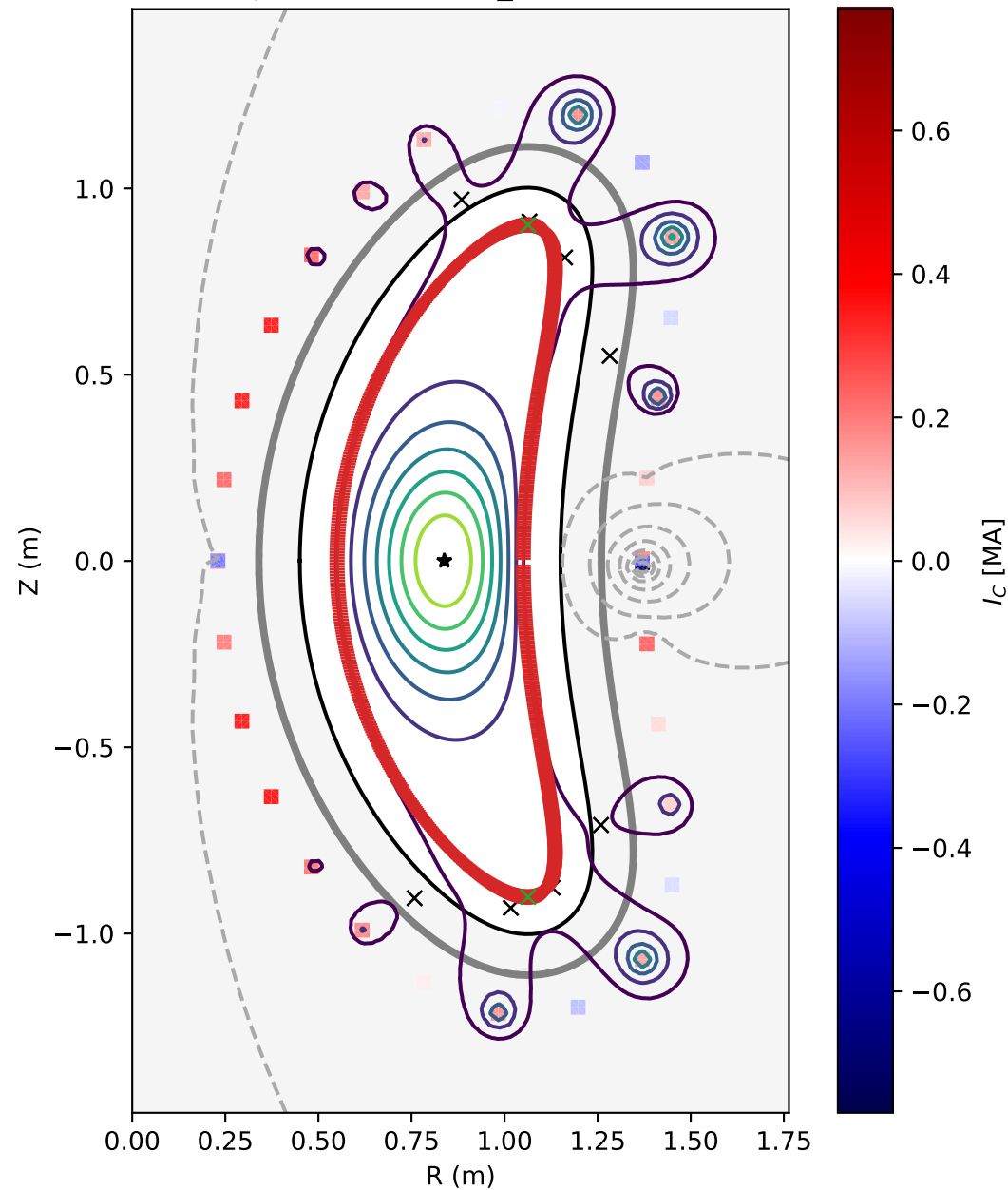
Plasma profiles:
pp_prof: $(1 - \psi^{1.421})^{2.464}$
ffp_prof: $(1 - \psi^{0.679})^{1.568}$

Scan params: B=2.5 C=14 G=9 H=3
alpha_pp=1.421 gamma_pp=2.464
alpha_ff=0.679 gamma_ff=1.568

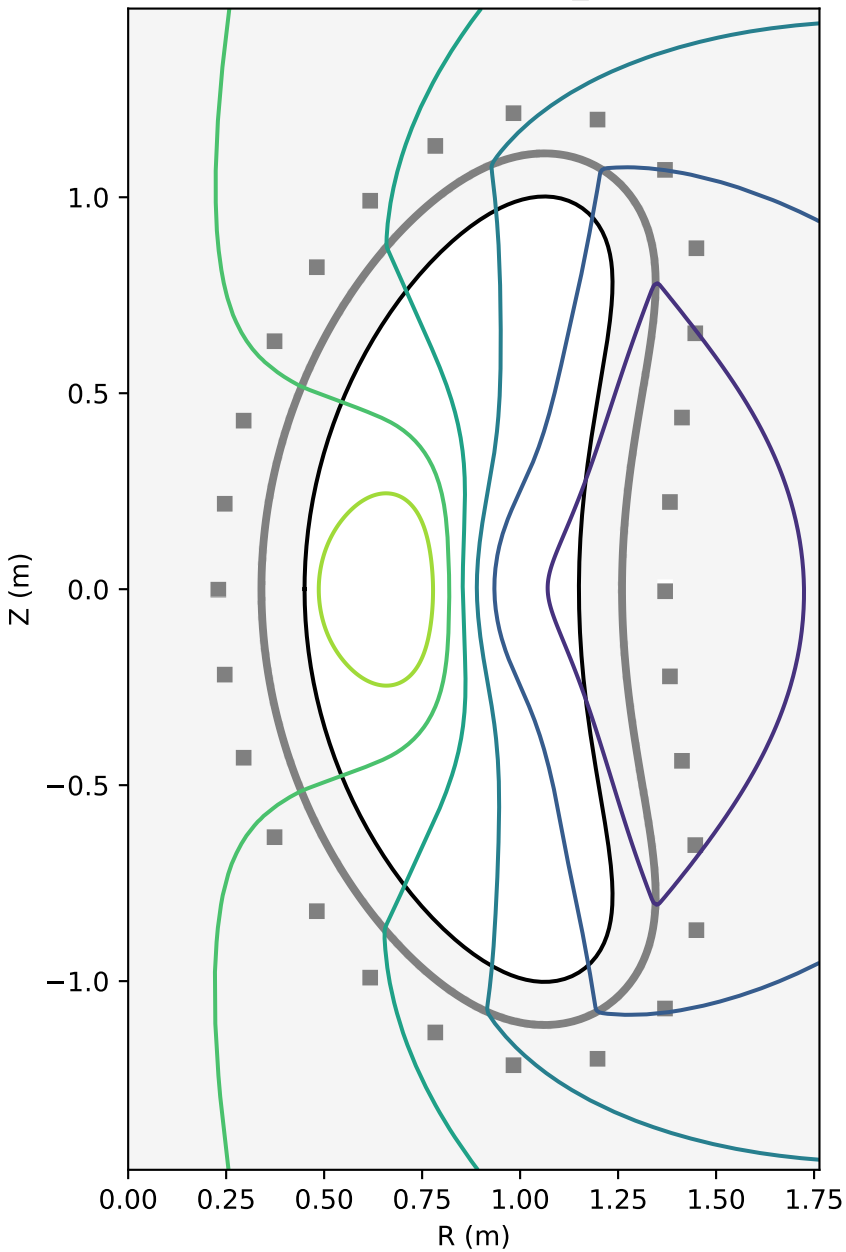
Geometry B=2.5, C=14, G=9, H=3, $\kappa=3.6$



Equilibrium flux $\kappa_{\text{actual}}=2.98$



Unstable eigenmode $\gamma\tau_w = -0.568$



Vertical Stability Analysis

Vertical Stability Results

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B=2.5 C=14 G=9 H=3

alpha_pp=1.421 gamma_pp=2.464

alpha_ff=0.679 gamma_ff=1.568

kappa_target = 3.600

kappa_actual = 2.980

beta_N = 1.055 %

growth rate (gamma) = -5.9841E+01 s-1

wall time (tau_w) = 9.4870E-03 s

gamma * tau_w = -0.5677

max coil current = 0.769 MA

Category: passive stable

axis_offset_pct = 4.79%

Equilibrium Statistics

Equilibrium Statistics:	
Topology	= Limited
Toroidal Current [A]	= 1.0000E+06
Current Centroid [m]	= 0.830 0.000
Magnetic Axis [m]	= 0.838 0.001
Elongation	= 2.980 (U: 2.974, L: 2.985)
Triangularity	= -0.654 (U: -0.657, L: -0.651)
Plasma Volume [m^3]	= 4.073
q_0, q_95	= 0.393 4.880
Peak Pressure [Pa]	= 3.1126E+05
Stored Energy [J]	= 1.5591E+05
<Beta_pol> [%]	= 60.5257
<Beta_tor> [%]	= 2.1532
<Beta_n> [%]	= 1.0546
Diamagnetic flux [Wb]	= 1.3771E-02
Toroidal flux [Wb]	= 1.3056E+00
l_i	= 1.7452